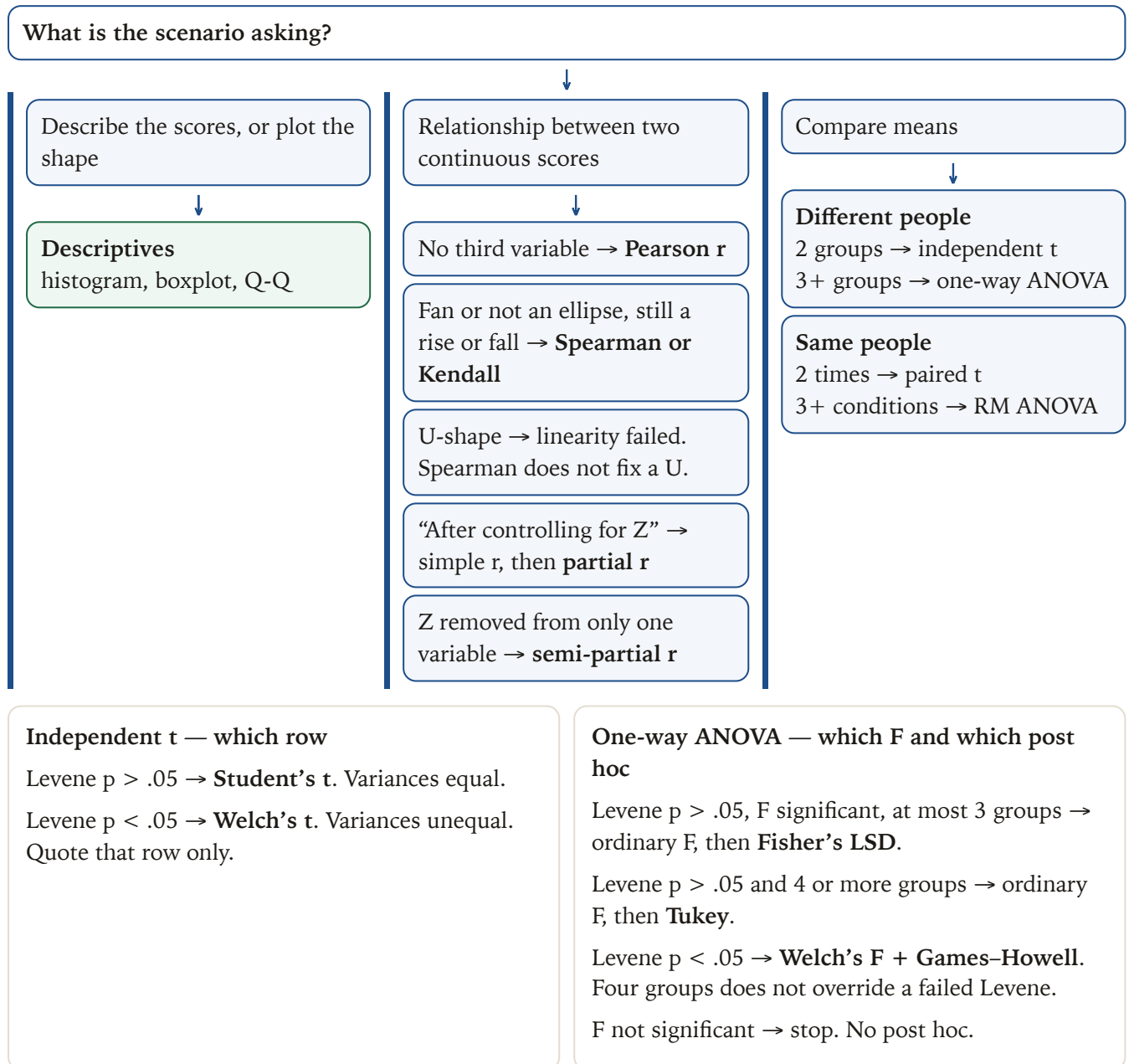


Lab Test 1 — which test, then click, read, write

28 Sep, CPD-3.41. One dataset, several scenarios. $\alpha = .05$. Outcome → Dependent Variable. Group label → Grouping Variable. ID column → never use it. Grouping variable measure type = Nominal. Outcome = Continuous.

Flowchart — decide the test first



RM ANOVA — which F

Mauchly $p > .05 \rightarrow$ uncorrected F.

Mauchly $p < .05$ and $\epsilon > .75 \rightarrow$ **Huynh–Feldt**.

Mauchly $p < .05$ and $\epsilon \leq .75 \rightarrow$ **Greenhouse–Geisser**.

Effect size is always **partial η^2** . Post hoc default is **Holm**.

Every p, every effect size

$p < .05 \rightarrow$ significant, reject H_0 . $p > .05$, including .051, .058, .150 $\rightarrow$ not significant, fail to reject H_0 . Still report the effect size. Do not write “there is no difference”.

jamovi prints $< .001 \rightarrow$ write $p < .001$.

Cohen’s d: 0.2 small, 0.5 medium, 0.8 large.

Between cutoffs, say “between ...”. $|d|$ below 0.2 is smaller than small.

$|r|$: $\geq .10$ small, $\geq .30$ medium, $\geq .50$ large. Sign is a separate word.

η^2 and partial η^2 have no small / medium / large cutoff. 0.71 means 71% of the variance. A 95% CI for a mean difference that includes 0 agrees with a non-significant t.

Descriptives and graphs

Click

Analyses → Exploration → Descriptives. Scores → **Variables**. A group label, only if you need a mean per group → **Split by**.

Statistics: N, Mean, Median, Mode, SD, Variance, Min, Max, Range, Skewness, Kurtosis, SE, Quartiles.

Plots: Histogram, Box plot, Q-Q. Bar plot only for a category.

Read	Means
Mean (SD)	First sentence of every later write-up. One pair per group.
Mean vs median	Mean above the median → right skew. Mean below the median → left skew.
Boxplot	Box = Q1 to Q3. Line = median. IQR = $Q3 - Q1$. Fence = $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$. Whisker stops at the last real score inside the fence. A point past the fence is an outlier.
Histogram	Bell = normal / Gaussian. One peak = one mode. Long right tail = positive skew.

The mean [score] was ____ (SD = ____), N = _____. The distribution was roughly symmetric / right-skewed / left-skewed (mean vs median, or the histogram). [If a boxplot:] Q1 = ____, median = ____, Q3 = _____. The upper fence was _____. The whisker stopped at _____. [score] was an outlier.

Independent-samples t-test

Use when one categorical variable has exactly two groups, different people, and the outcome is continuous. Three or more groups → ANOVA. Same people twice → paired t. Two continuous scores → correlation.

Click

Analyses → T-Tests → Independent Samples T-Test.

- Dependent Variables = the outcome.
- Grouping Variable = the two-group label.
- Tick **Student's** and **Welch's**.
- Hypothesis: Group 1 $\neq$ Group 2, unless the question states a direction.
- Additional: Mean difference, Confidence interval, Cohen's d, Descriptives.
- Assumption checks: Homogeneity test (Levene), Q-Q plot. Shapiro-Wilk over-calls non-normality. Prefer the Q-Q plot.

Read	Indicator
Both means and SDs	Higher mean = the direction.
Q-Q	Points hug the line → normality met. A bend, or ends peeling off → not met. One or two end points in a small sample are not enough to call a violation.
Levene F(df), p	$p > .05$ → report Student's. $p < .05$ → report Welch's. The Welch df will not be $N - 2$.
t(df), p, 95% CI, d	From one row only. $p < .05$ and the CI excludes 0 → the means differ. $d = .74$ is between medium and large.

The mean [outcome] in [group 1] was __ (SD = __), whereas the mean in [group 2] was __ (SD = __). Normality of errors was checked with a Q-Q plot of the residuals and was met / not met. Homogeneity of variance was checked with Levene's test, $F(_, _) = _, p = _,$ and was met / violated. [Student's t because Levene was not significant. / Welch's t because homogeneity was violated.] There was a significant / non-significant difference, $t(_) = _, p = _,$ 95% CI [__, __], Cohen's $d = _ (size),$ such that [group] had higher [outcome].

Class: Anastasia 74.53 (SD 9.00) vs Bernadette 69.06 (SD 5.77). Levene $F(1, 31) = 2.49, p = .125,$ so Student's. $t(31) = 2.12, p = .043,$ CI [0.20, 10.76], $d = .74,$ between medium and large. Anastasia higher.

Paired-samples t-test

Use **when** the same people are measured twice. Three or more conditions → RM ANOVA. Different people → independent t. No grouping variable, no Levene, no Welch.

Click

Analyses → T-Tests → Paired Samples T-Test. Put the two measures in one pair. Student's only. Hypothesis: Measure 1 $\neq$ Measure 2. Additional: mean difference, CI, Cohen's d, descriptives. Assumptions: Normality, Q-Q.

Read	Indicator
Mean and SD of each time	Higher mean = the direction.
Q-Q	Hug the line → normality of the difference scores is met.
t(df), p, CI, d	df = people - 1. t(19) means 20 people. CI includes 0 → not significant. d uses 0.2 / 0.5 / 0.8. d = 1.45 is large.

The mean of [time 1] was __ (SD = __), whereas the mean of [time 2] was __ (SD = __). Normality of errors was checked with a Q-Q plot and was met / not met. A paired-samples t-test showed a significant / non-significant difference, t(__) = __, p = __, 95% CI [__, __], Cohen's d = __ (size), such that scores were higher at [time].

Class: test 1 = 56.98 (6.62), test 2 = 58.38 (6.41). t(19) = 6.48, p < .001, CI [0.95, 1.86], d = 1.45, large. Second test higher.

Correlation — Pearson, Spearman, Kendall

Use **when** both variables are continuous and you are not comparing group means. A third variable to hold constant → partial, after this simple r .

Click

Analyses → Regression → Correlation Matrix. Both variables in the box. **Pearson** first. Hypothesis: Correlated (two-tailed) unless the question states a direction. Tick report significance, N, confidence interval.

Assumption pictures: Analyses → Regression → Linear Regression. Dependent = Y. Covariates = X. Tick Q-Q plot of residuals and Residual plots. This is only to judge the pictures. Do not write a regression answer.

Read	Indicator
Scatter, shape	Straight cloud → linearity met. U or a clear curve → Pearson is the wrong model. Spearman also misses a U.
Scatter, cloud	One ellipse → bivariate normality met. Two clumps → violated.
Fitted vs residuals	Even vertical spread → homoscedasticity met. Fan → violated, so report Spearman or Kendall and say why. Wider in the middle → say it is ambiguous, pick a side, give the reason. A cloud that trends up or down is failed linearity (bias), not a fan.
r , p	Sign = direction. $ r $ gives small / medium / large. $p = .051$ is not significant, and a medium r is still reported.
Time-linked rows	Independence fails. Name autocorrelation (AR, MA, ARMA, ARIMA are the names). Do not use Pearson.

There was a significant / non-significant [positive / negative] [small / medium / large] relationship between [X] and [Y], $r = ___$, $p = ___$. As [X] increased, [Y] increased / decreased. Independence holds by design / is violated because the rows are time-linked. The scatterplot was linear / curved. The cloud was / was not one ellipse. Homoscedasticity was met / violated on the residual plot (say why). [If you switched:] Spearman's rho / Kendall's tau-b = $___$, $p = ___$, because [normality / homoscedasticity] failed.

Class: sleep and attention, $N = 100$, $r = .35$, $p = .051$. Positive, medium, not significant. Fail to reject H_0 . The relationship is not “absent”. Dan: baby's sleep $r = .63$; grumpiness $r = -.90$; baby with grumpiness $r = -.57$. All $p < .001$, all large. Negative means the second score goes down as the first goes up.

Partial and semi-partial correlation

Partial removes Z from both X and Y. Use it when the question says “after controlling for” or “holding Z constant”.

Semi-partial removes Z from only one variable, usually the predictor. The two off-diagonal cells can differ. Name the cell. Partial r is the same either way and is usually larger.

Click

Run the Correlation Matrix first, so you have the simple r. Then Analyses → Regression → Partial Correlation. Variables = the two you care about. Control Variables = Z. Pearson. Correlation type: **Partial** or **Semipartial**. The default is Partial. Report significance.

Compare simple r with the controlled r	Name
Weaker, still significant	Partial confounder
No longer significant	Confounder
Stronger	Suppressor
Sign flips	Negative suppressor

The simple relationship between [X] and [Y] was [direction] and [size], $r = ___$, $p = ___$. After controlling for [Z], the partial correlation was $r = ___$, $p = ___$. The relationship weakened / was no longer significant / strengthened / flipped direction. This pattern is a partial confounder / confounder / suppressor / negative suppressor.

Class: stress and exam $r = .664$, $p < .001$, positive large. After daily study time, partial $r = -.618$, $p < .001$, negative large. The sign flipped. Negative suppressor.

One-way independent ANOVA

Use **when** one grouping variable has 3 or more groups and different people are in each group. Two groups → independent t. Same people → RM ANOVA.

Click

Analyses → ANOVA → One-Way ANOVA. Dependent Variable = outcome. Grouping Variable = the groups. Leave “Assume equal (Fisher’s)” on. If Levene fails, also use “Don’t assume equal (Welch’s)” and quote Welch.

Descriptives. Assumption checks: Homogeneity test, Q-Q. Post hoc only after you have seen Levene and the omnibus F.

η^2 : if One-Way ANOVA has no effect-size box, also open Analyses → ANOVA → ANOVA. Dependent = outcome. Fixed Factors = that one grouping variable only. Tick η^2 . Do not add a second factor. Fisher’s F should match. Copy η^2 from this table. In a one-way design partial η^2 prints the same number. Write η^2 .

Read	Indicator
Every group’s mean and SD	Missing these is a wrong write-up.
Q-Q	Hug the line → normality met.
Levene	$p > .05$ → Fisher’s F. $p < .05$ → Welch’s F.
$F(df1, df2), p$	Omnibus. $p < .05$ → the means are not all equal. It does not name the pair. $df1 = k - 1$, $df2 = N - k$.
η^2	Proportion of variance. No Cohen band.
Post hoc	See the flowchart. Report every pair, including $p > .05$. Post hoc df is the residual df.

Means (SDs): [A] __ (__), [B] __ (__), [C] __ (__). Homogeneity of variance was met / violated, Levene’s $F(_, _) = _, p = _$. Normality of errors was checked with a Q-Q plot and was met / not met. A one-way independent ANOVA [or Welch’s F] showed a significant / non-significant effect of [IV] on [DV], $F(_, _) = _, p = _, \eta^2 = _$. Post hoc [Fisher’s LSD / Tukey / Games–Howell]: [pair], $t(_) = _, p = _$. In words: [who differed, and who did not].

Class, drugs: Levene $F(2, 15) = 1.45, p = .266$, met, so Fisher’s and LSD. $F(2, 15) = 18.61, p < .001, \eta^2 = 0.71$ (71%). Joyzepam > anxifree, $t(15) = 4.36, p < .001$. Joyzepam > placebo, $t(15) = 5.88, p < .001$. Anxifree vs placebo, $t(15) = 1.52, p = .150$, not significant. The slide once printed that last p as < .001.

One-way repeated-measures ANOVA

Use **when** the same people do 3 or more conditions. Two conditions → paired t. Different people → one-way independent ANOVA. Effect size is always partial η^2 .

Click

Analyses → ANOVA → Repeated Measures ANOVA. Name the factor (Condition). Set Levels. Label them. Drag each measure into its level. There is no grouping variable.

Effect size: **partial η^2** . In a one-way RM, η^2 and generalised η^2 print the same number as each other. That number is not partial η^2 . Partial η^2 ignores between-person variance. Tick partial η^2 .

Assumption checks: Sphericity tests (Mauchly), Q-Q. Post hoc: put the factor across and tick **Holm**, unless the question names another correction. Least conservative → most: None, Tukey, Holm, Bonferroni, Scheffé.

Read	Indicator
Mean and SD of every condition	The pattern of means is the direction.
Q-Q	Hug the line → normality met.
Mauchly $\chi^2(df)$, p	$p > .05$ → uncorrected F. For 3 conditions the df is usually 2. Copy the df jamovi prints.
ϵ	Only if Mauchly fails. $\epsilon > .75$ → Huynh–Feldt. $\epsilon \leq .75$ → Greenhouse–Geisser. Copy df from that row.
F, p, partial η^2	$p < .05$ → conditions differ somewhere.
Post hoc	A significant F can sit next to every pair $p > .05$. Say so. Do not crown a winner.

[A] M = __ (SD = __), [B] M = __ (SD = __), [C] M = __ (SD = __). Mauchly's test indicated that sphericity was met / violated, $\chi^2(_) = _, p = _$. [If violated: $\epsilon = _$. Because ϵ was above / not above .75, Huynh–Feldt / Greenhouse–Geisser was used.] Normality of errors was checked with a Q-Q plot and was met / not met. There was a significant / non-significant effect of condition, $F(_, _) = _, p = _, \text{partial } \eta^2 = _$. Post hoc Holm: [pairs with t, df, p]. If no pair is significant, write that the omnibus test was significant but no pairwise comparison reached .05.

Class: Speech 7.17 (1.47), Conceptual 6.17 (1.47), Syntax 4.33 (1.63). Mauchly $\chi^2 = 0.85$, $p = .720$, met, so the uncorrected F $F(2, 10) = 6.39$, $p = .013$, partial $\eta^2 = .581$. Tukey pairs $p = .381, .058, .102$. Omnibus significant, no pair at .05. Speech vs Syntax at .058 is not significant.